



DEALER BEST PRACTICE SERIES

Mining Trucks Dump Body
Removal & Install System

Maintenance and Repair Application Component Life Management Component Renewal (CRC) MARC Management

Mining Trucks Dump Body Removal & Install System0

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1.0 Introduction

The demand for productivity and cost optimization has pushed the OHT manufacturers to address bigger hauling capacity, which subsequently has increased the size of the equipment and components. These bigger equipment and components has to be maintained representing a significant challenge to the maintenance & repair operation in terms of facilities, special tooling, people skills and safety. In this scenario the removal and installation of the dump body of a mining truck represent a task that requires special care because of its size and weight. This task requires extensive coordination of resources and observation of safety procedures to minimize the inherent risk of working with a suspended load of such weight and dimensions.

Historically the dump body removal and installation has been executed by the use of two high capacity portable cranes that are operated in precise coordination guided by a crew of normally a minimum of 5 skilled people per crane plus 2 low boy truck operators for the transportation of the dump body. In some specific cases it has been noticed that as much as 7 to 14 persons are being assigned to this operation.



Illustration 1: Pictures of a typical dump body removal / installation operation executed with the assistance of two portable cranes.

This operation has to be done in an open and leveled area and in some cases can last more than 30 hours. Therefore, the service crew has to pay special attention to all safety procedures to minimize the inherent risks of working with a “suspended” load that in some cases can be over 50 metric tons.

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It is a risky and unpredictable operation, because of being performed in an open area can be adversely affected by the weather conditions, especially wind conditions that can delay or stop the execution of it for as long as it is necessary to operate within the safety recommendations. Normally a wind speed of 35 km/h (21.7mph) is accepted as the maximum limit for maneuvering this type of loads with a portable crane.

The following bullets summarize the main concerns or disadvantages of the traditional “suspended load” dump body removal and install procedure:

- Method or concept used. Suspended load of as much as 52 plus tons (797 Dump Body case)
- Accessories. Use of lifting devices such as cables/chain/rope that represent a high risk if not maintained, inspected and installed / used carefully.
- Service crew is forced to execute the task of removing the dump body and hoist cylinder pins under a suspended load.
- Availability of suitable portable crane to handle the dump body weight. This normally requires extensive coordination to ensure the availability of the cranes when they are needed. A perfect coordination won't ensure 100% crane availability when for example a plant emergency arises.
- Weather conditions. It can adversely affect the dump body R&I operation, delaying or cancelling completely the task. This in fact can ruin any possible coordination done in advance. As a consequence, the maintenance planning cannot be achieved.
- Manpower. Significant manpower is required to complete the operation. Normally the people involved in this operation can be as much as 7(using one crane) to 14 persons (using two cranes). 1 lead supervisor, 1 safety expert, 2 or 4 mechanics to control the position cables (“wind control” cables), 2 or 4 Riggers to assist the crane operator by signals, 1 or 2 skilled crane operators and 2 low boy truck operators.
- Execution time. Because of all the concerns and uncontrolled events that could be present at the time of the operation, the dump body execution time can be as high as 24 hours, and in some cases higher. Normally the planning department schedules the dump body change for 2 eight hours shift, but this estimate can be easily affected as explained before by many external factors.
- Pin removal. The pins removal operation can be also a challenge for the service crew when using the two cranes method, because of the inability of controlling the dump body position in the four (4) lifting points separately. The pin removal can be in some cases a bottle neck in terms of execution time.
- Dump body transportation costs. The uncontrolled factors already mentioned may in some cases increase the transportation costs when dump body repairs are done outside the mine site, e.g. parking costs

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The Best Practice herein presented represents a significant improvement, an innovative solution to minimize or eliminate the risks and disadvantages of the traditional dump body removal & install operation.



Illustration 2: Picture of the OHT Dump Body Removal & Installation System

The OHT Dump Body Removal & Install System eliminates the risks and disadvantages of working with a suspended load by the use of an hydraulically operated system that consist mainly in four (4) support arms located in 4 respective columns (single zone configuration) that lift, maneuver and position the dump body as needed by remote control. This approach changes the operation from a “suspended load” to a “supported load” scenario eliminating or minimizing the disadvantages mentioned previously.

The OHT Dump Body R&I System is a solution created and manufactured by Novamine SpA. (Patents involved: CL20080002638 and CL20080002637; Patent Pending: CL20070001876), a specialized company in the field of development and manufacturing of innovative mining solutions located in Antofagasta Chile.

2.0 Best Practice Description

The OHT Dump Body Removal & Install System consist of an adjustable structure of 6 columns (full configuration) operated hydraulically that are positioned to form 2 work zones dedicated to perform the operation of dismounting and mounting the truck dump body.

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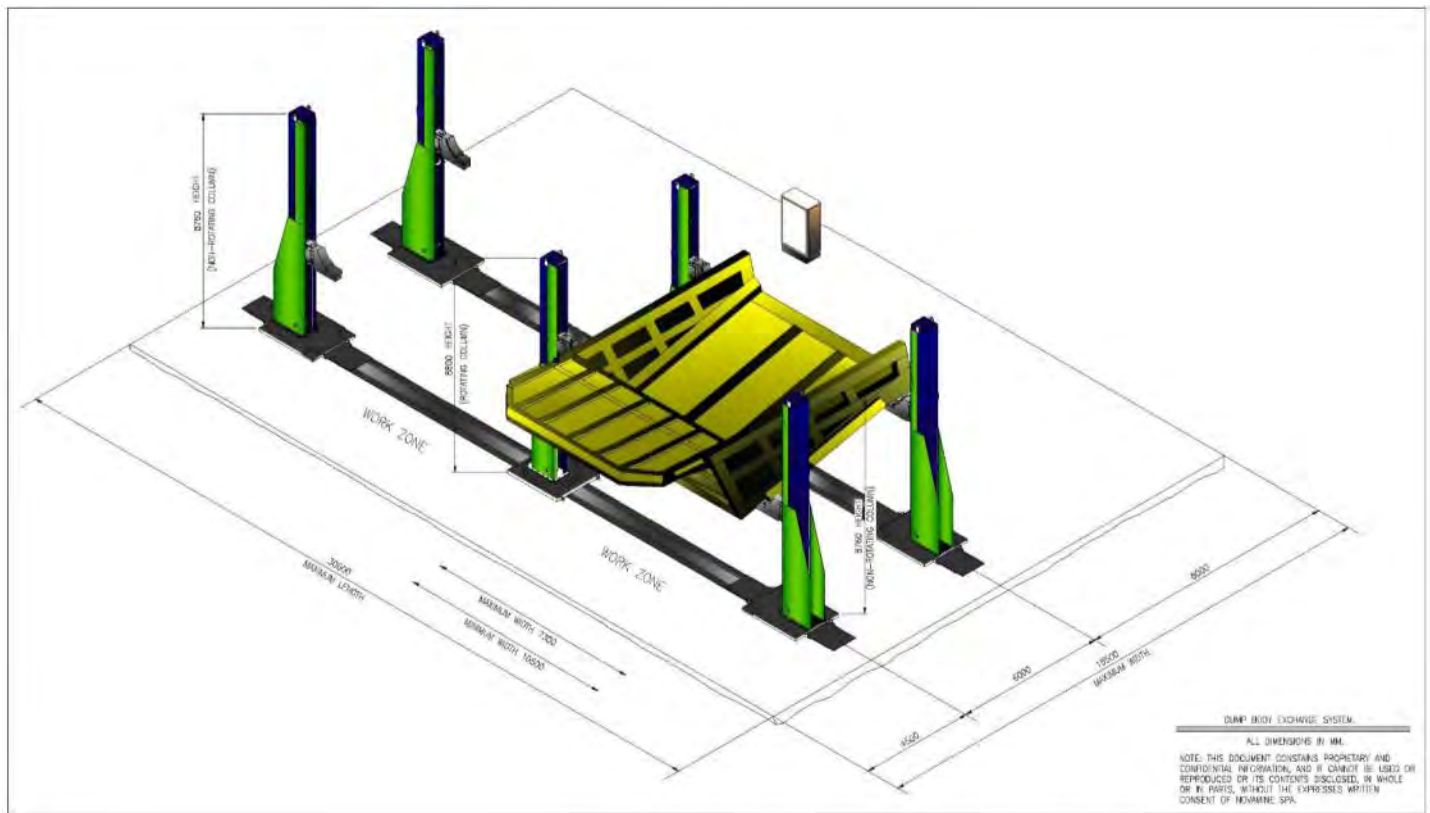


Illustration 3: Dump Body R&I System diagram

The system operates basically with 4 support arms installed in 4 respective columns that combined provides tridimensional adjustment (x, y, z). The sideways or horizontal adjustment of the columns is designed to facilitate the positioning (entry and exit) of the truck. And the vertical adjustment of the support arms are dedicated to lift and maneuver the body to remove it from the truck and to positioning the dump body in 4 support fixtures or directly onto a low boy transportation on highway truck.

The horizontal movement of the columns is handle by the operator from the low part of each column and the vertical movement of the arms is operated either through wireless remote control or panel control located in the OTH Dump Body R&I System.

Note: The central columns rotate in 180 degrees to allow the operation in both work zones.

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Illustration 4: Columns and support arms schematic

The operation starts with the positioning of the truck in the center of the work zone. To facilitate the proper location of the truck positioning marks are being added to the work zone floor. Once the truck has been positioned the columns and support arms are adjusted to support the body.

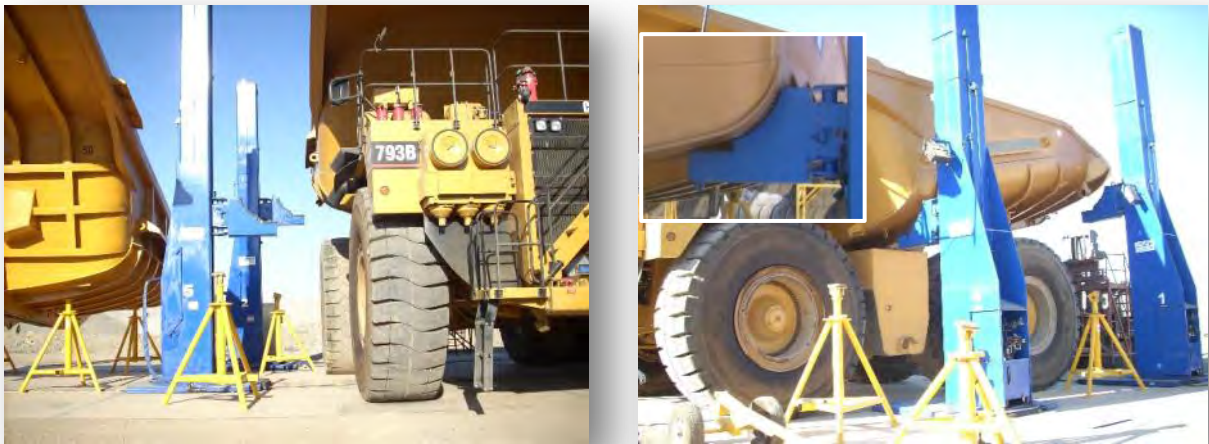


Illustration 5: The truck is located in the center of the work zone. Support arms and columns adjusted to support the dump body.

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After the dump body is supported by the arms, the dump body gets lifted enough to remove the load from the pins and ensure the easy removal of them. To facilitate the pins removal the 4 support arms can be adjusted independently.

Once the pins are removed the dump body can be lifted and the truck can be released.



Illustration 7: Dump Body removed and the truck released. The system is operated through a wireless remote control.

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Next the dump body can be loaded into a low boy truck or mounted on supports to perform a local repair.



The general technical specifications of the system are the following:

- Columns hoist capacity (per column): 20 Tons.
- Total hoist capacity: 80 Tons.
- Distance between columns (width): 6 meters.
- Truck models capable of servicing: from Cat 785 to Cat 797 size range.
- Hoist displacement: 5.25 meters
- Lift time: 8 minutes.
- Maximum Hydraulic Pressure: 3.000 Psi.
- Electrical power supply: 380 V, 50 – 60 Hz.
- Electric Motor Power: 10 HP.
- Fully functional wireless remote control.
- Support fixtures (support dump body).
- Dump body download from low boy truck System.

3.0 Implementation Steps

The implementation steps to follow are related to the preparation of the work zones, building the work zones platforms (concrete floor) and to follow the recommendations of the manufacturer.

The details for the implementation of the system can be obtained directly by contacting the manufacturer or the mining operations listed below that are currently operating the system.

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The manufacturer contact information is:

NOVAMINE SPA.

Street: Av. Antonio Rendic 6904

Postal Code 1265823

Antofagasta – Chile

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EDITOR'S NOTE: Although Caterpillar does not endorse this particular supplier, Finning Chile has experience with the following products or companies. We welcome additional supplier recommendations from our dealer network.

Mining Operations that are currently operating the system:

Mining Operation	Parental Company	Location	Since	Website
Mantos Blancos	Anglo American	Antofagasta II Region – Chile	2003	http://www.angloamerican-chile.cl
Codelco - Radomiro Tomic	Codelco	Calama II Region – Chile	2004	http://www.codelco.com
El Abra	Freeport-McMoRan Copper & Gold	Calama II Region – Chile	2006	http://www.fcx.com
El Tesoro	Antofagasta Minerals SA.	Antofagasta II Region – Chile	2009	http://www.tesoro.cl
Candelaria	Freeport-McMoRan Copper & Gold	Antofagasta II Region – Chile	2009	http://www.fcx.com
Codelco – Chuquicamata	Codelco	Calama II Region – Chile	2012	http://www.codelco.com

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Codelco - Andina	Codelco	Los Andes V Region - Chile	2013	http://www.codelco.com
Los Pelambres	Antofagasta Minerals SA.	Salamanca IV Region - Chile	2013	http://www.pelambres.com
Escondida	BHP Billiton Ltd.	Antofagasta II Region - Chile	2013	http://www.bhpbilliton.com
Codelco – Ministro Hales	Codelco	Calama II Region – Chile	2013	http://www.codelco.com

4.0 Benefits

The main benefits aimed by the designer / manufacturer and documented by the customers currently using the system are:

- Significant improvement in safety by moving from a “suspended load” to a “supported load” operation. The risk reduction using the body R&I system can be observed in many locations:
 - o Working conditions under a heavy suspended load eliminated.
 - o No need for operating mobile cranes eliminating the risk of potential erroneous coordination between both lifting machines.
 - o No need for lifting devices (Cables, slings,...), eliminating the risk of faulty material and inadequate installation of these devices.
 - o Improved visibility for the person in charge of the operation.
 - o Risk associated to the pins removal mitigated.
 - o Mechanical locks prevent /minimize the risk of a dump body falling in the case of a malfunction of the hydraulic hoist system.
 - o Unexpected weather changes influences eliminated.
- In terms of the Maintenance & Repair Processes, the benefits are also significant in terms of efficiency, resources, and effectiveness. The following are the main benefits documented:
 - o Dump Body R&I time reduce significantly. Execution time reduction of as much as 10 times has been documented.
 - o Positive influence in MR (Maintenance Ratio) by a significant reduction of the personnel needed to complete the dump body R&I operation. Normally 3 to 4 trained people is sufficient to perform the task.
 - o No need for the use of cranes eliminating the cost and coordination of highly demanded equipment.
 - o Unexpected weather related delays eliminated contributing effectively to comply with maintenance event planning.

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- o No additional specialized equipment needed for a different truck model or size. The system is capable of handling various dump body sizes and configurations.
- o Eliminate extra costs associated with the dump body transportation system by reducing the unexpected delays.
- o Improved certain maintenance and repair tasks efficiency and effectiveness by removing the dump body to expedite the execution of the tasks such as R&I of transmissions, torque converters, final drives, etc. The need for special support equipment is minimized and the use of the shop overhead crane maximized.

5.0 Resources Required

Apart from the acquisition of dump body R&I system the main resources needed are related to the assignment and preparation of the area in where this equipment will be installed.

All technical specifications, such as the ones for the concrete floor of the work areas, need to be discussed with the manufacturer of the equipment.

The manufacturer contact information is:

NOVAMINE SPA.

Street: Av. Antonio Rendic 6904

Postal Code 1265823

Antofagasta – Chile

Phone: +56-55-2566234/ +56-55-2566229

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6.0 Supporting Attachments / References

Explanatory videos from NOVAMINE SPA. The first video is in Spanish with English subtitles and it shows the customer's opinion regarding the benefits of the Dump Body R&I System. The second video is in English and it explains very effectively the basic operation of the Dump Body R&I System.

- <https://vimeo.com/80365293>
- <https://www.youtube.com/watch?v=24pYDsIPRcY>

7.0 Related Best Practices

N/A

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8.0 Acknowledgements

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